1] Test the effect of 3 different fertilizers (A, B, C) on crop yield (kg). The yields obtained are.

Fertilizer A	Fertilizer B	Fertilizer C
20	25	22
22	27	24
21	26	23
23	28	25

Hypotheses:
- H_0 (NULL): $\mu_1 = \mu_2 = \mu_3$ (all means are equal)
- H_1 (Alternative): Atleast one ~~meaning~~ mean is different.

Sol^n Here we use 1-way ANOVA

Step 1: Calculate Means.

Mean A : $(20 + 22 + 21 + 23)/4 = 21.5$
Mean B : $(25 + 27 + 26 + 28)/4 = 26.5$
Mean C : $(22 + 24 + 23 + 25)/4 = 23.5$
Grand Mean (Total average) : $(20 + 22 + 21 + 23 + 25 + 27 + 26 + 28 + 22 + 24 + 23 + 25)/12 = 23.83$

Step 2: Calculate Sum of Squares between (SS_between)
Formula : $\Sigma\, n * (\text{Group mean} - \text{Grand Mean})^2$ for each group.

Group A : $4 * (21.5 - 23.83)^2 = \cancel{186.5956}\ 21.7156$
Group B : $4 * (26.5 - 23.83)^2 = \cancel{7.1289}\ 28.5156$
Group C : $4 * (23.5 - 23.83)^2 = \cancel{0.1089}\ 0.4356$
SS_Between $= 21.7156 + 28.5156 + 0.4356$
$= 50.7907$

Date :

Step 3 : Calculate Sum of Squares within (SS-within)
formula : Σ (Datapoint - Group Mean)^2

Group A : $(20-21.5)^2 + (22-21.5)^2 + (21-21.5)^2 + (23-21.5)^2 = 5$
Group B : $(25-26.5)^2 + (27-26.5)^2 + (26-26.5)^2 + (28-26.5)^2 = 5$
Group C : $(22-23.5)^2 + (24-23.5)^2 + (23-23.5)^2 + (25-23.5)^2 = 5$
SS-Within $= 5 + 5 + 5 = 15$.

Step 4 : Degrees of freedom (df)

df-between : (Number of groups -1) = $(3-1) = 2$
df-within : (Total data points - Number of groups) $= 12 - 3 = 9$

Step 5 : Calculate Mean Squares (MS) Average Variation

MS-between : SS-between / df-between $= 50.7907/2 = 25.3953$
MS-within : SS-within / df within $= 15/9 = 1.6667$

Step 6 : The F-Statistics.

F = MS-between / MS-within $= 25.3953/1.6667 = 15.23$

Step 7 : Critical value & Conclusion.

- F-Critical $(2, 9, \alpha = 0.05) \approx 4.26$
+ since $F = 15.23 > 4.26$, reject H_0

There is a significant difference in Crop yields among the three fertilizers.

Date :

2] Test whether the mean processing time (minutes) differs for three machines.

Machine 1	Machine 2	Machine 3
12	10	8
14	11	9
13	12	10

Soln Hypotheses.

- H_0 (NULL) : $\mu_1 = \mu_2 = \mu_3$ (all means are equal)
- H_1 (Alternate) : Atleast one mean is different.

Step 1 : Calculate mean.

Mean A : (12 + 14 + 13) / 3 = 13
Mean B : (10 + 11 + 12) / 3 = 11
Mean C : (8 + 9 + 10) / 3 = 9
Grand Mean (Total Average) : (12 + 14 + 13 + 10 + 11 + 12 + 8 + 9 + 10) ÷ 9 = 11

Step 2 : Calculate sum of squares between (SS-between)
Formula : $n * (\text{Group Mean} - \text{Grand Mean})^2$ for each group

Group A : $3 * (13 - 11)^2 = 12$
Group B : $3 * (11 - 11)^2 = 0$
Group C : $3 * (9 - 11)^2 = 12$

SS-Between = 12 + 0 + 12 = 24

Date :

Step 3 : Calculate Sum of Squares within (SS-within)

formula : Σ (Data point - Group Mean)^2

Group A : $(12-13)^2 + (14-13)^2 + (13-13)^2 = 2$

Group B : $(10-11)^2 + (11-11)^2 + (12-11)^2 = 2$

Group C : $(8-9)^2 + (9-9)^2 + (10-9)^2 = 2$.

SS-within = $2 + 2 + 2 = 6$.

Step 4 : Degrees of freedom (df)

df-between : (Number of groups - 1) = $(3-1) = 2$

df-within : (Total data points - # of groups) = $(9-3) = 6$.

Step 5 : Calculate Mean Squares (MS) Average variation.

MS-between : SS-between / df-between = $24/2 = 12$

MS-within : SS-within / df-within = $6/6 = 1$

Step 6 : The F-statistics.

F = MS-between / MS-within = $12/1 = 12.0$.

Step 7 : Critical Value and conclusion.

- F-critical $(2, 6, \alpha = 0.05) \approx 5.14$
- since $F = 12 > 5.14$, reject H_0

There is a significant difference in mean processing time among the three machines ($p < 0.05$)

3] A researcher studies the effect of factor A and factor B on student test scores.
→ Factor A : A1. Traditional A2. Online (Teaching method)
→ Factor B : B1. Low B2. High. (study time)

The observed scores are:

Teaching Method	Low Study time (B1)	High Study time (B2)
Traditional (A1)	60, 62	70, 72
Online (A2)	65, 67	75, 78

Test at $\alpha = 0.05$ whether:
1. Teaching method has a significant effect.
2. Study time has significant effect.
3. There is an interaction effect.

Sol^n Here we use Two-Way-ANOVA

Step 1 : Calculate Means (cellwise)

Traditional ~~Mean~~ Low : 60, 62 (Mean = 61)
Traditional High : 70, 72 (Mean = 71)

Online Low: 65, 67 (Mean = 66)
Online High: 75, 78 (Mean = 76.5)

Step 2 : Calculate factor Means.

Traditional Mean = (61 + 71)/2 = 66
Online Mean = (66 + 76.5)/2 = 71.25

Hypotheses.

Factor A: Teaching Method
$H_0 A$: $\mu_{A1} = \mu_{A2}$
$H_1 A$: Means are diff

Factor B : Study time.
$H_0 B$: $\mu_{B1} = \mu_{B2}$
$H_1 B$: Means are diff

Interaction (A × B)
$H_0 AB$: No interaction effect
$H_1 AB$: Interaction exists.

Date :

low study time mean $= (61 + 66)/2 = 63.5$

high study time mean $= (71 + 76.5)/2 = 73.5$

Grand Mean $= (61 + 71 + 66 + 76.5)/4 = 68.625$

Average Table

	L	H	Avg
T	61	71	66
O	66	76.5	71.25
Avg	63.5	73.5	68.625

count table

2	2	4
2	2	4
4	4	8

Step 3 : Calculate Sum of squares.

Formula : $\sum n * (\text{Group mean} - \text{Grand mean})^2$

factor A (Teaching method) $= 4\left[(66 - 68.625)^2 + (71.25 - 68.625)^2\right]$

$\boxed{SS\_A = 55.125}$

factor B (Study time) $= 4\left[(63.5 - 68.625)^2 + (73.5 - 68.625)^2\right]$

$\boxed{SS\_B = 210.125}$

$SS\_AB = \sum\sum n\,(\text{Cell mean} - \text{Row mean} - \text{Column mean} + \text{Grand mean})^2$

$= 2\left[(61 - 66 - 63.5 + 68.625)^2 + (71 - 66 - 73.5 + 68.625)^2 + (66 - 71.25 - 63.5 + 68.625)^2 + (76.5 - 71.25 - 73.75 + 68.625)^2\right]$

$= 2\left[(0.125)^2 + (-0.125)^2 + (-0.125)^2 + (0.125)^2\right]$

$= 8\,(0.125)^2 = 0.125.$

$\boxed{SS\_AB = 0.125}$

$SS\_Error = \sum(\text{Individual observation} - \text{cell mean})^2$

$= \left[(60-61)^2 + (62-61)^2 + (70-71)^2 + (72-71)^2 + (65-66)^2 + (67-66)^2 + (75-76.5)^2 + (78-76.5)^2\right]$

$= [1 + 1 + 1 + 1 + 1 + 1 + 2.25 + 2.25]$

$\boxed{SSE = 10.5}$

Step 4: Calculate degree of freedom

df_A = No of levels of A − 1 = 2 − 1 = 1
df_B = No of levels of B − 1 = 2 − 1 = 1
df_AB = df_A × df_B = 1 × 1 = 1

Total observation = (No of levels of A)(No of levels of B)(Observation per cell) = (2)(2)(2) = 8

df_Error = (No of levels of A)(No levels of B) × (observation per cell − 1)
= (2)(2)(2 − 1)
= 4

df_total = Total observation − 1 = 7

Step 5: Source table calculation.

Source	SS	df	MS	F statistics
Factor A (method)	55.125	1	55.125	21 (1, 4)
Factor B (Study time)	210.125	1	210.125	80.0476 (1, 4)
Interaction	0.125	1	0.125	0.0476 (1, 4)
Error	10.5	4	2.625	
Total	275.875	7	39.4107	

MS_A = SS_A / df_A
MS_B = SS_B / df_B
MS_AB = SS_AB / df_AB
MS_Error = SS_Error / df_Error

F_A = MS_A / MS_Error
F_B = MS_B / MS_Error
F_AB = MS_AB / MS_Error

Step 6 : Conclusions ($\alpha = 0.05$, F- Critical $(1,4) \approx 7.71$)

Source	F_Calculated	F_Critical.	Decision.
Teaching Method Effect	21.	7.71	Fcalc > Fcritical, Reject H_0
Study Time Effect	80.0476	7.71	Fcalc > Fcritical, Reject H_0
Interaction Effect.	0.0476	7.71	Fcalc < Fcritical, fail to reject H_0

- Teaching method has significant effect on the test scores
- Study time has significant effect on the test scores.
- No interaction effect between teaching method & study time. The effect of teaching method does not depend on study time level.